EXAMPLE 16.4

The products of combustion of an unknown hydrocarbon $C_x H_y$ have the following composition as measured by an Orsat apparatus:

$$CO_2\ 8.0\%,\ CO\ 0.9\%,\ O_2\ 8.8\% \text{ and } N_2\ 82.3\%$$

Determine: (a) the composition of the fuel, (b) the air-fuel ratio, and (c) the percentage excess air used.

Solution Let a moles of oxygen be supplied per mole of fuel. The chemical reaction can be written as follows.

$$C_xH_y + a\,O_2 + 3.76\,a\,N_2$$

$$\rightarrow 8\,CO_2 + 0.9\,CO + 8.8\,O_2 + 82.3\,N_2 + \frac{y}{2}\,H_2O$$

Oxygen balance gives

$$2a = 16 + 0.9 + 17.6 + \frac{y}{2} \tag{1}$$

Nitrogen balance gives

$$3.76a = 82.3$$

$$\therefore \quad a = 21.89$$

By substituting the value of a in Eq. (1),

$$y = 18.5$$

Carbon balance gives

$$x = 8 + 0.9 = 8.9$$

Therefore, the chemical formula of the fuel is $C_{8.9}H_{18.5}$. The composition of the fuel is

$$\%\text{ carbon} = \frac{8.9 \times 12}{8.9 \times 12 + 18.5 \times 1} \times 100 = 85.23\%$$

$\therefore$ % hydrogen = 14.77%

$$\text{(b) Air fuel ratio} = \frac{32\,a + 3.76\,a \times 28}{12\,x + y}$$

$$= \frac{21.89\,(32 + 3.76 \times 28)}{12 \times 8.9 + 18.5}$$

$$= \frac{3005}{125.3} = 24$$

(c) % Excess air used

$$= \frac{8.8 \times 32}{21.89 \times 32 - 8.8 \times 32} \times 100 = 67.22\%$$

EXAMPLE 16.5

Determine the heat transfer per kg mol of fuel for the following reaction

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O\ (l)$$

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The reactants and products are each at a total pressure of 100 kPa and 25°C.

Solution By the first law

$$Q_{c.v.} + \sum_R n_i \bar{h}_i = \sum_P n_e \bar{h}_e$$

From Table C in the appendix

$$\sum_R n_i \bar{h}_i = (\bar{h}_f)_{CH_4} = -74{,}874 \text{ kJ}$$

$$\sum_P n_e \bar{h}_e = (\bar{h}_f^0)_{CO_2} + 2(\bar{h}_f^0)_{H_2O\,(l)}$$

$$= -393{,}522 + 2(-285{,}838)$$

$$= -965{,}198 \text{ kJ}$$ Ans.

$\therefore$

$$Q_{c.v.} = -965{,}198 - (-74{,}873)$$

$$= -890{,}325 \text{ kJ}$$ Ans.

EXAMPLE 16.6

A gasoline engine delivers 150 kW. The fuel used is C_8H_{18} (*l*) and it enters the engine at 25°C. 150% theoretical air is used and it enters at 45°C. The products of combustion leave the engine at 750 K, and the heat transfer from the engine is 205 kW. Determine the fuel consumption per hour, if complete combustion is achieved.

Solution The stoichiometric equation (Fig. Ex. 16.6) gives

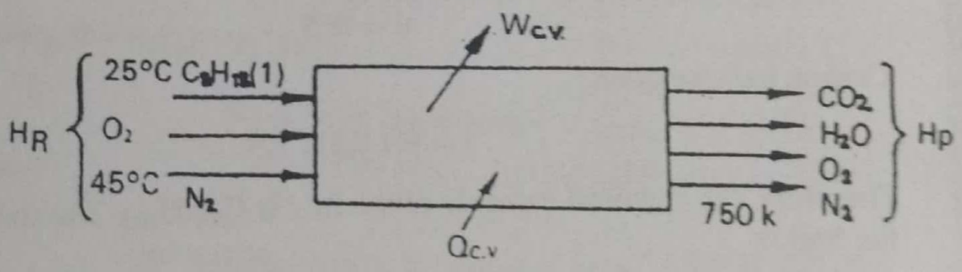

FIG. EX. 16.6

$$C_8H_{18}\,(1) + 12.5\,O_2 + 3.76 \times 12.5\,N_2$$

$$\rightarrow 8CO_2 + 9H_2O + 3.76 \times 12.5\,N_2$$

With 150% theoretical air

$$C_8H_{18} + 18.7O_2 + 70\,N_2$$

$$\rightarrow 8CO_2 + 9H_2O + 6.25O_2 + 70\,N_2$$

$$H_R = \sum_R n_i [\bar{h}_f^0 + (\bar{h}_T^0 - \bar{h}_{298})]_i$$

$$= 1(\bar{h}_f)_{C_8H_{18}(l)} + 18.7[\bar{h}_f^0 + (\bar{h}_T - \bar{h}_{298})]_{O_2}$$

$$+ 70[\bar{h}_f^0 + (\bar{h}_T - \bar{h}_{298})]_{N_2}$$

$$= (-249{,}952) + 18.7 \times 560 + 70 \times 540$$

$$= -201{,}652 \text{ kJ/kgmol fuel}$$

$$H_P = 8\,[-393{,}522 + 20288] + 9\,[-241{,}827 + 16087] + 6.25\,[14171] + 70\,[13491]$$

$$= 8\,(-373{,}234) + 9\,[-225{,}740] + 6.25\,(14171) + 70\,[13491]$$

$$= -3{,}884{,}622 \text{ kJ/kg mol fuel}$$

Energy output from the engine, $W_{C.V.} = 150$ kW

$$Q_{C.V.} = -205 \text{ kW}$$

Let n kgmol of fuel be consumed per second.
By the first law

$$n\,(H_R - H_P) = W_{C.V.} - Q_{C.V.}$$

$$n\,[-201{,}652 + 3{,}884{,}662] = 150 - (-205) = 355 \text{ kW}$$

$$\therefore \quad n = \frac{355 \times 3600}{3683010} \text{ kgmol/h}$$

$$= 0.346 \text{ kgmol/h}$$

$\therefore$ Fuel consumption rate

$$= 0.346 \times 114 = 39.4 \text{ kg/h}$$

Ans.

EXAMPLE 16.7

Determine the adiabatic flame temperature when liquid octane at 25°C is burned with 300% theoretical air at 25°C in a steady flow process.

Solution The combustion equation with 300% theoretical air is given by

$$C_8H_{18}(1) + 3(12.5)O_2 + 3(12.5)(3.76)\,N_2 \rightarrow$$

$$\rightarrow 8CO_2 + 9H_2O + 25O_2 + 141N_2$$

By the first law

$$H_R = H_P$$

$$\sum_R n_i[\bar{h}_{f0} + \Delta\bar{h}]_i = \sum_P n_e[\bar{h}_{f0} + \Delta\bar{h}]_e$$

Now

$$H_R = (\bar{h}_{f0})_{C_8H_{18}(1)} = -249{,}952 \text{ kJ/kgmol fuel}$$

$$H_P = \sum_P n_e[\bar{h}_{f0} + \Delta\bar{h}]_e = 8\,[-393{,}522 + \Delta\bar{h}_{CO_2}] + 9\,[-241{,}827 + \Delta\bar{h}_{H_2O}] + 25\,\Delta\bar{h}_{O_2} + 141\,\Delta\bar{h}_{N_2}$$

The exit temperature, which is the adiabatic flame temperature, is to be computed by trial and error satisfying the above equation.

Let $T_e = 1000$ K, then

$$H_P = 8(-393{,}522 + 33{,}405) + 9(-241{,}827 + 25{,}978) + 25(22.707) + 141(21{,}460)$$

$$= -1226{,}577 \text{ kJ/kg mol}$$

If $T_e = 1200$ K

$$H_P = 8(-393{,}522 + 44{,}484) + 9(-241{,}827 + 34{,}476) + 25(29{,}765) + 141(28{,}108)$$

$$= +46{,}537 \text{ kJ/kg mol}$$

If $T_e = 1100$ K,

$$H_P = 8(-393{,}522 + 38{,}894) + 9(-241{,}827 + 30{,}167) + 25(26{,}217) + 141(24{,}757)$$

$$= -595{,}964 \text{ kJ/kg mol}$$

Therefore, by interpolation, the adiabatic flame temperature is 1182 K. Ans.

EXAMPLE 16.8

(a) Propane (g) at 25°C and 100 kPa is burned with 400% theoretical air at 25°C. Assume that the reaction occurs reversibly at 25°C, that the oxygen and nitrogen are separated before the reaction takes place (each at 100 kPa, 25°C), that the constituents in the products are separated, and that each is at 25°C, 100 kPa. Determine the reversible work for this process. (b) If the above reaction occurs adiabatically, and each constituent in the products is at 100 kPa pressure and at the adiabatic flame temperature, compute (a) the increase is entropy during combustion, (b) the irreversibility of the process, and (c) the availability of the products of combustion.

Solution The combustion equation (Fig. Ex. 16.8) is

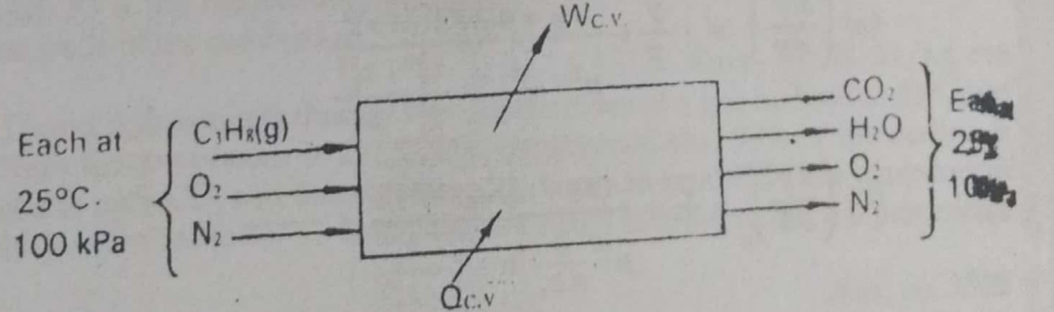

FIG. EX. 16.8

$$C_3H_8(g) + 5\,(4)\,O_2 + 5(4)(3.76)\,N_2 \rightarrow$$

$$3\,CO_2 + 4\,H_2O\,(g) + 15\,O_2 + 75.2\,N_2$$

(a) $W_{rev} = \sum_R n_i \bar{g}_i - \sum_P n_e \bar{g}_e$

From Table 16.1

$$W_{rev} = (\bar{g}_{f0})_{C_3H_8(g)} - 3(\bar{g}_{f0})_{CO_2} - 4(\bar{g}_{f0})_{H_2O(g)}$$

$$= -23{,}316 - 3(-394{,}374) - 4(-228{,}583)$$

$$= 2{,}074{,}128 \text{ kJ/kg mol}$$ Ans.

$$= \frac{2{,}074{,}128}{44.097} = 47{,}035.6 \text{ kJ/kg}$$ Ans.

(b) $H_R = H_P$

$$(\bar{h}_{f^0})_{C_3H_8(g)} = 3(\bar{h}_{f^0} + \Delta\bar{h})_{CO_2} + 4(\bar{h}_{f^0} + \Delta\bar{h})_{H_2O(g)} + 15\Delta\bar{h}_{O_2} + 75.2\Delta\bar{h}_{N_2}$$

From Table 16.1

$$-103{,}847 = 3(-393{,}522 + \Delta\bar{h})_{CO_2} + 4(-241{,}827 + \Delta\bar{h})_{H_2O(g)} + 15\Delta\bar{h}_{O_2} + 75.2\Delta\bar{h}_{N_2}$$

Using Table C in the appendix, and by trial and error, the adiabatic flame temperature is found to be 980 K.

The entropy of the reactants

$$S_R = \sum_R (n_i \bar{s}_i^0)_{298} = (\bar{s}^0_{C_3H_8(g)} + 20\bar{s}^0_{O_2} + 75.2\bar{s}^0_{N_2})_{298}$$

$$= 270.019 + 20(205.142) + 75.2(191.611)$$

$$= 18{,}782.01 \text{ kJ/kgmol K}$$

The entropy of the products

$$S_P = \sum_P (n_e \bar{s}_e^0)_{980} = (3\bar{s}^0_{CO_2} + 4\bar{s}^0_{H_2O(g)} + 15\bar{s}^0_{O_2} + 75.2\bar{s}^0_{N_2})_{980}$$

$$= 3(268.194) + 4(231.849) + 15(242.855) + 75.2(227.485)$$

$$= 22{,}481.68 \text{ kJ/kgmol K}$$

$\therefore$ The increase in entropy during combustion

$$S_P - S_R = 3{,}699.67 \text{ kJ/kg mol K}$$ Ans.

The irreversibility of the process

$$I = T_0[\sum_P n_e \bar{s}_e - \sum_R n_i \bar{s}_i]$$

$$= 298 \times 3699.67$$

$$= 1{,}102{,}501.66 \text{ kJ/kgmol}$$

$$= \frac{1{,}102{,}501.66}{44.097} = 25{,}001 \text{ kJ/kg}$$

The availability of combustion products

$$\psi = W_{rev} - I$$

$$= 47{,}035.6 - 25{,}001$$

$$= 22{,}034.6 \text{ kJ/kg}$$ Ans.